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Cholesterol and POPC segmental order parameters in lipid membranes: solid state ^{1}H–^{13}C NMR and MD simulation studies†

Tiago Mendes Ferreira,*[a] Filipe Coreta-Gomes,[b] O. H. Samuli Ollila,[a] Maria João Moreno,[b] Winchil L. C. Vaz[b] and Daniel Topgaard[a]

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The concentration of cholesterol in cell membranes affects membrane fluidity and thickness, and might regulate different processes such as the formation of lipid rafts. Since interpreting experimental data from biological membranes is rather intricate, investigations on simple models with biological relevance are necessary to understand the natural systems. We study the effect of cholesterol on the molecular structure of multi-lamellar vesicles (MLVs) composed of 1-palmitoyl-2-oleoyl-*sn-glycero*-3-phosphocholine (POPC), a phospholipid ubiquitous in cell membranes, with compositions in the range 0–60 mol% cholesterol. Order parameters, $|S_{CH}|$, are experimentally determined by using ^{1}H–^{13}C solid-state nuclear magnetic resonance (NMR) spectroscopy with segmental detail for all parts of both the cholesterol and POPC molecules, namely the ring system and alkyl chain of the sterol, as well as the glycerol backbone, choline headgroup and the *sn*-1 and *sn*-2 acyl chains of POPC. With increasing cholesterol concentration the acyl chains gradually adopt a more extended conformation while the orientation and dynamics of the polar groups are rather unaffected. Additionally, we perform classical molecular dynamics simulations on virtual bilayers mimicking the POPC–cholesterol MLVs investigated by NMR. Good agreement between experiments and simulations is found for the cholesterol alignment in the bilayer and for the $|S_{CH}|$ profiles of acyl chains below 15 mol% cholesterol. Deviations occur for the choline headgroup and glycerol backbone parts of POPC, as well as for the phospholipid and cholesterol alkyl chains at higher cholesterol concentrations. The unprecedented detail of the NMR data enables a more complete comparison between simulations and experiments on POPC–cholesterol bilayers and may aid in developing more realistic model descriptions of biological membranes.

Introduction

The preferred synthesis of cholesterol in mammalian cells over other sterols has been argued as an evolutionary step linked to the optimization of membrane cellular processes.[1–3] Mammals have an intricate homeostasis mechanism allowing precise control of cholesterol levels in their membranes.[4] Failure of these homeostatic processes may lead to severe disease, *e.g.* in humans with the genetic disease familial hypercholesterolemia, cholesterol levels rise manyfold above normal, leading to a higher probability for heart attacks.[5,6] Cholesterol levels depend on cell and organelle type, being for instance around 17.5% of total dry weight in myelin but almost absent in mitochondrial membranes,[7,8] and ranging from 30–50 mol% of the total lipid fraction in plasma membranes.[9] The composition of biological membranes in terms of other lipid molecules than cholesterol and of membrane proteins also varies between cell and organelle types. Such differences are most likely related to the optimization of distinct functions on different membranes but a clear understanding of the reason for varying concentrations has not yet been established.

One aspect which is not fully rationalized is the dependence of membrane protein functionality on membrane lipid composition.[10] In some cases it may be understood by specific interactions between lipids and proteins while in other cases also the physical properties of the lipid bilayer might play a significant role.[10,11] The presence of cholesterol in biological

[a] *Physical Chemistry, Lund University, P.O. Box 124, SE-221 00 Lund, Sweden. E-mail: tiago.ferreira@fkem1.lu.se*

[b] *Departamento de Química, Faculdade de Ciências e Tecnologia da Universidade de Coimbra (FCTUC), Largo D. Dinis, Rua Larga, 3004-535 Coimbra, Portugal*

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membranes is probably relevant in both cases. First of all, it increases the order of the lipid molecules as well as the thickness and stiffness of the bilayer. One consequence directly related to this effect is the decrease of the membrane's permeability to small molecules or ions. It has also been suggested, for instance, that the elastic properties of the bilayer may affect rhodopsin function[12] and that the increase in thickness of the endoplasmic reticulum membrane leads to the loss of activity of sarco(endo)plasmic reticulum ATPase.[13] On the other hand, the analysis of membrane protein extracts from cells indicates that cholesterol, or lipid molecules affected by cholesterol, might bind directly to specific proteins (ref. 10 and references therein). Furthermore, cholesterol is believed to play an important role in the formation of lipid rafts, *i.e.* postulated microdomains involved in the cell membrane biosynthetic and endocytic traffic.[14,15] Aside from the importance of cholesterol on the structural properties of membranes, it is also a precursor in the biological synthesis of a number of vital compounds such as bile salts, steroid hormones or vitamin D, to name a few.[16]

A detailed understanding of lipid bilayer properties is necessary to fully understand the biophysical issues discussed above. Here we report a study on bilayers consisting of mixtures with varying compositions of two biologically ubiquitous molecules, 1-palmitoyl-2-oleoyl-*sn-glycero*-3-phosphocholine (POPC) and cholesterol. Even though real biological membranes are substantially more complex, with several different kinds of molecules, these model systems are useful to understand basic effects of cholesterol on a cell membrane-like environment. We perform united-atom MD simulations of POPC–cholesterol bilayers, by using a commonly used united-atom model,[17] together with R-type proton-detected local field (R-PDLF) NMR experiments[18] on POPC–cholesterol multi-lamellar vesicles (MLVs). Atomistic (or nearly atomistic) MD simulation techniques currently provide the most detailed molecular descriptions available on model membrane systems, and NMR is a powerful and non-invasive technique to determine detailed properties of model membranes with atomic resolution. These two techniques complement each other since MD simulations enable the interpretation of NMR data to be more comprehensive while, on the other hand, quantitative comparisons between MD and NMR observables help to optimize the force-fields used in the simulations.

R-PDLF NMR spectroscopy is a recent alternative over the more conventional ^{2}H NMR methods to measure order parameters, S_{CH}, for individual C–H bonds,

$$S_{CH} = \tfrac{1}{2}\langle 3\cos^2\theta - 1\rangle, \quad (1)$$

where θ is the angle between the direction of the C–H bond and the bilayer normal. In this work we show that by performing R-PDLF NMR on samples with natural abundance of isotopes it is possible to determine $|S_{CH}|$ values for almost all C–H segments in POPC–cholesterol bilayers, both from POPC and cholesterol molecules simultaneously. These experimental measurements show that while the phospholipid acyl chain tails reorient towards having more *trans* conformations, the orientation and dynamics of the polar groups are rather unaffected through cholesterol concentrations in the range of 0–60 mol%. The highly selective and detailed information from R-PDLF NMR experiments enables a significantly better comparison between the simulation model and experimental results than previously.[19–23] Our results show that the ordering of acyl chains induced by cholesterol is very well described by the MD model at low cholesterol concentration. On the other hand, the choline headgroup, glycerol backbone, and acyl chains at higher concentrations, give different C–H order parameters than the NMR experiments; and the independence of the polar headgroup structure of POPC on cholesterol concentration is not captured. Moreover, a comparison of experimental and MD simulation cholesterol order parameters shows that while the ring structure is well described by simulations, this is not the case for the flexible alkyl chain of the sterol.

We expect that the results and discussion given here – pointing out a number of molecular details on the POPC–cholesterol bilayer structure and confronting the simulations to the new collection of NMR data – will aid in the development of more realistic model descriptions of biological membranes.

Materials and methods

Preparation of POPC–cholesterol MLVs

Cholesterol, 1-palmitoyl-2-oleoyl-phosphatidylcholine (POPC) and POPC deuterated at the *sn*-1 chain (POPC-d31) was from Avanti Polar Lipids (Alabaster, AL). Chloroform and methanol were purchased from Sigma. All reagents used were of the highest commercially available purity. Aqueous suspensions of lipids were prepared as described previously in ref. 24. The multi-lamellar dispersions were then centrifuged, the supernatants were discarded and the pellets were transferred to Bruker 4 mm HR-MAS rotors with a sample volume of 12 μl. Phospholipid concentrations were determined through a modified version of the Bartlett phosphate assay[25] and cholesterol concentrations were determined by the Lieberman–Burchard method as described in ref. 26. Absorption spectra were recorded on a Unicam UV530UV/Vis spectrophotometer. For samples with higher cholesterol content, the cholesterol/POPC ratio was calculated from ^{13}C direct polarization experiments (90° pulse and acquisition with TPPM decoupling[27] on the ^{1}H channel, details below). The quantification was done by integrating the peaks at 13.9 ppm and 12.2 ppm in the ^{13}C spectrum, corresponding to the methyl groups of the acyl chains of POPC and to one of the methyl groups in the cholesterol ring structure respectively. The concentrations studied were as follows: 0%, 7%, 15%, 34%, 50% and 60% of cholesterol (mol%) in the total amount of POPC + cholesterol lipid moles. Two independent samples were used for measurements at each concentration.

R-PDLF, INEPT and CP ^{1}H–^{13}C NMR experiments

The measurements were performed using a Bruker Avance AVII-500 NMR spectrometer operating at a ^{1}H frequency of

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500.23 MHz equipped with a standard bore 4 mm CP-MAS HX probe. All experiments were done with the sample under magic angle spinning (MAS) at the spinning frequency of 5 kHz. The separated local field (SLF) experiment R-PDLF (ref. 18 and 28 for more details) was used with refocused INEPT[29,30] or CP[31] as the polarization transfer method. The following setup was used: spectral width of 149.5 ppm; refocused INEPT with τ_1 and τ_2 of 1.79 ms and 1.19 ms respectively; CP contact time equal to 700 μs; radiofrequency pulses set to give the nutation frequencies: 45.00 kHz ($R18_1^7$ pulses), 63.45 kHz (^{13}C 90° and 180° pulses), 63.45 kHz (^{1}H INEPT pulses), 50 kHz (^{1}H decoupling pulses), 50.76–63.45 kHz (^{1}H CP ramp pulse during contact time), 53.66 kHz (^{13}C CP pulse during contact time); t_1 with increments equal to 399.6 μs (0% colesterol, 7% and 15% cholesterol) and 132.2 μs (34% colesterol, 50% and 60% cholesterol); for each t_1 value, 256 scans were recorded during an acquisition time of 100 ms using 5 s recycle delay. MAS and RF heating effects were checked by using a rotor with methanol.[32] A difference of 1.55 ppm between the two peaks of the ^{1}H NMR spectrum of methanol was measured before and immediately after performing the longest of all R-PDLF experiments, giving a constant effective temperature of 300 K.

After performing a 2D spectral processing scheme to the time domain results as described in ref. 28, one gets a 2D spectrum with a number of splittings at different positions. The two dimensions of the spectrum correspond to the ^{13}C chemical shift that gives resolution for the different segments (direct dimension) and to the dipolar field recoupled during t_1 from which the order parameters may be calculated (indirect dimension). The size of the splittings, $\Delta\nu_{CH}$, at the different chemical shifts is proportional to the effective dipolar coupling, d_{CH}, by $\Delta\nu_{CH} = 0.315\ d_{CH}$, which in turn is proportional to the order parameter magnitude, $|S_{CH}|$ (eqn (1)), through,

$$d_{CH} = \frac{\hbar}{2}\left(\frac{\mu_0}{4\pi}\right)\gamma_H\gamma_C\langle r^{-3}\rangle|S_{CH}|, \qquad (2)$$

with μ_0 being the vacuum permittivity, γ_H and γ_C the gyromagnetic constants for the ^{1}H and the ^{13}C nuclei and $\langle r^{-3}\rangle$ the average inverse cubic length for a CH chemical bond. We assume a ^{1}H–^{13}C rigid lattice value d_{CH}^{max} = 21.5 kHz (for $|S_{CH}| = 1$).[33]

MD simulation setup of the POPC–cholesterol bilayers

Molecular dynamics simulations were run by using the GROMACS 4.5.3 simulation package.[34–37] For POPC we used the same force-field as in ref. 38 which is based on the Berger parametrization for lipids,[39] except that dihedrals next to double bonds are described as in ref. 40. A GROMOS based force-field for cholesterol was taken from ref. 41. The use of these force-fields is a common procedure in several previous simulations (ref. 17 and references therein). The molecule types in the cholesterol force-field were changed from CH_2/CH_3 to LP_2/LP_3 to avoid overcondensation of the bilayer as suggested in ref. 42. All the initial structures were bilayers – consisting of two leaflets, each with 64 lipid molecules, and both having the same composition – hydrated with at least 3000 water molecules. Snapshots for two of the systems simulated are shown in Fig. 1. Energy minimization *via* the steepest descent method was done for all initial structures.

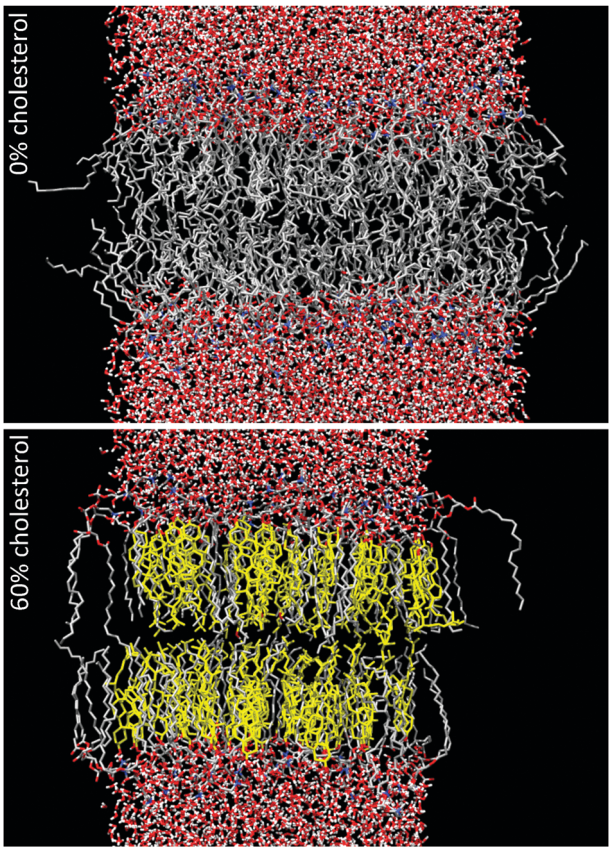


Fig. 1 MD simulation snapshots of a POPC bilayer and a POPC–cholesterol bilayer after 20 ns of equilibration. The colors identify different atoms/molecules: red is used for oxygen, white for hydrogens, blue for nitrogen, and grey and yellow denote the carbons from POPC and cholesterol respectively. The view scale used to generate the pictures was the same. Periodic boundary conditions were used in all directions.

All simulations were carried out with the Berendsen thermostat and barostat[43] with pressure and temperature coupling constants of 1.0 ps and 0.1 ps respectively. The phospholipid, cholesterol and water molecules were separately coupled to a heat bath at 300 K, and the semi-isotropic pressure coupling was applied separately in the *xy*-direction (bilayer plane) and the *z*-direction (bilayer normal) with a coupling constant of 1 bar. Cutoff radius for van der Waals interactions was set to 1.0 nm. Particle mesh Ewald (PME) summations[44] were applied for long-range electrostatic interactions with a grid spacing of 0.12 nm and a cutoff radius of 1.0 nm was employed for real space summation. The time step was 2 fs, using LINCS[45] to constrain all bond lengths of lipids, whereas the SETTLE[46] algorithm was used for the SPC water.[47] The atoms coordinates were saved each 10 ps. The total time for simulations was 100 ns for each system from which only the last 80 ns were used for calculating averages.

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C–H order parameters were calculated by first generating the hydrogen locations for all molecules in each time configuration. Then we use the equation

$$S_{CH} = \frac{1}{2NM}\sum_{m=1}^{M}\sum_{n=1}^{N}\left(\frac{3r_{nm,z}^{\ 2}}{|\vec{r}_{nm}|^2} - 1\right), \qquad (3)$$

where $\vec{r}_{nm}$ is the C→H vector of the C–H bond at molecule m and time frame n, with coordinates relative to an axis coordinate frame which has the z-axis parallel to the bilayer normal. The sum goes through all M molecules and N time frames. Note that in some cases the order parameter can be different for the different hydrogens connected to the same carbon. We calculate and report order parameters separately for all hydrogens. This discrimination is important since the two separate values can often be measured from experiments as well.

Results

Assignment of the ^{13}C chemical shifts and complete $|S_{CH}|$ profile of the phospholipid bilayer

A R-PDLF spectrum of POPC MLVs at full hydration is shown in Fig. 2 (for an explanation of the spectrum see the last paragraph of NMR experiments in Materials and Methods). Assignments for the glycerol, choline and some of the acyl chain ^{13}C peaks can be found in the literature,[48] but there is no previous detailed assignment of the several peaks in the crowded spectral region between 29 and 31 ppm. To determine the complete $|S_{CH}|$ profile from 2D (chemical shift–dipolar coupling) NMR spectroscopy such ^{13}C assignments must also be performed. In order to identify the previously unassigned ^{13}C peaks, we first distinguish the *sn*-1 (palmitoyl) peaks from the *sn*-2 (oleoyl) peaks. This distinction is done by performing R-PDLF experiments for two different samples of POPC MLVs in excess water: one having natural abundance of isotopes and another where the phospholipids have a fully deuterated *sn*-1 chain (POPC-d31). At the bottom of Fig. 2 we show

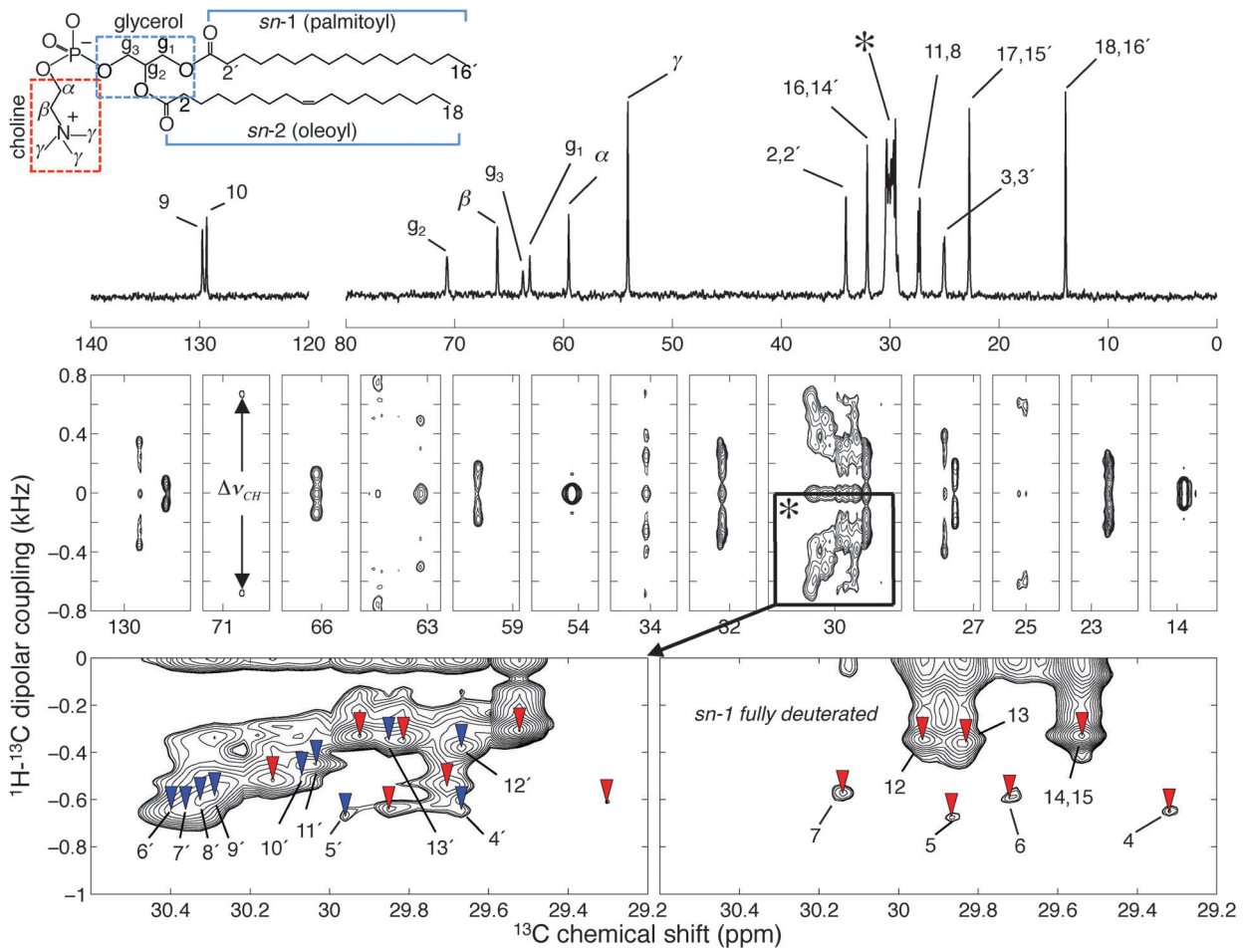


Fig. 2 2D ^{1}H–^{13}C R-PDLF spectrum of fully hydrated POPC multi-lamellar vesicles recorded at 300 K and B_0 = 11.74 T with INEPT polarization transfer, 5 kHz magic-angle spinning, 50 kHz TPPM ^{1}H decoupling during ^{13}C acquisition, and 45 kHz R18$_1^7$ ^{1}H–^{13}C dipolar recoupling in the indirect dimension. The peak for the CH_3 carbons C_{18} and $C_{16'}$ is used as chemical shift reference at 13.9 ppm. Peak assignments as reported by Volke *et al.*[48] (*) Previously unassigned crowded spectral region around 30 ppm shown in detail for POPC (bottom, left) and POPC-d31 (bottom, right) multi-lamellar vesicles. Peaks from the *sn*-1 (palmitoyl) and *sn*-2 (oleoyl) chains are labeled with blue and red triangles respectively. The order parameters of the distinct carbons can be calculated from their splittings, $\Delta\nu_{CH}$.

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magnifications of the R-PDLF spectra for POPC and POPC-d31 at the crowded spectral regions. The *sn*-1 chain ^{13}C signals in the deuterated sample are not present since the source of magnetization transfer from protons to carbons was removed. Likewise, the bottom right spectra in Fig. 2 only shows peaks from the *sn*-2 chain. By comparing these spectra it is then possible to identify which peaks belong to the *sn*-1 and *sn*-2 chains. The next step is performing the peak assignment for each of these sets of palmitoyl and oleoyl segments. Our choice here is made such that, for these previously unassigned ^{13}C peaks, the $|S_{CH}|$ profile mimics the profile predicted from the MD simulation. This identification based on MD is done exclusively for the previously unassigned ^{13}C peaks and solely for the sample without cholesterol. The initial ^{13}C NMR assignment is then kept consistently throughout the study.

The complete order parameter profiles obtained from both experiments and simulations are shown in Fig. 3. For comparison we also show previously measured order parameters for *sn*-1 chain,[49] *sn*-2 chain[49,50] and for the polar headgroup region.[51] Note that, when it comes to simulations, we always show the individual order parameters from the two distinct hydrogens in a CH_2 group, while from the NMR experiments we can often only measure one order parameter from each methylene. If only one point is visible from simulations (red dots) it means that the two points overlap.

Cholesterol $|S_{CH}|$ profile

At higher concentrations of cholesterol it is also possible to determine a large number of cholesterol ^1H–^{13}C dipolar splittings from CP based R-PDLF experiments as shown in Fig. 4. The efficiency of the CP polarization transfer depends on both the order parameter and correlation time, τ_c, for C–H reorientation.[52] Since τ_c is most likely the same for both hydrogens in the same CH_2 group, the C–H pair with a higher $|S_{CH}|$ value will have a higher CP signal,[52] hence obscuring the signal of its complementary C–H pair. Thus, although two distinct ^{13}C–^1H dipolar couplings are expected for each CH_2 segment at the cholesterol ring structure, due to their rigid nature, it is only possible to measure directly the dipolar coupling corresponding to the highest $|S_{CH}|$ value. We resolve unambiguously order parameters for all the CH_2, CH_3 and C–H

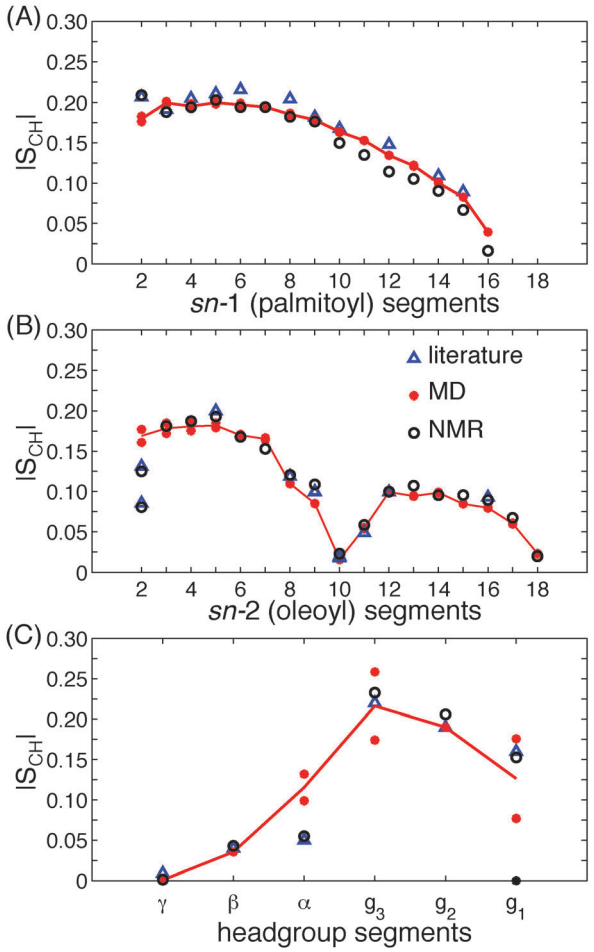


Fig. 3 Order parameter magnitude $|S_{CH}|$ *vs.* carbon segment number for the *sn*-1 and *sn*-2 acyl chains of POPC (A and B respectively); and for the choline and glycerol backbone of the POPC headgroup (C). Data from fully hydrated POPC at 300 K obtained with ^1H–^{13}C solid-state NMR (black dots) and MD simulations (red dots), as well as literature data (blue triangles) from ^2H NMR (*sn*-1[49] and *sn*-2[49,50] at 300 K) and ^1H–^{13}C NMR (headgroup in samples of egg yolk lecithin at 305 K[51]).

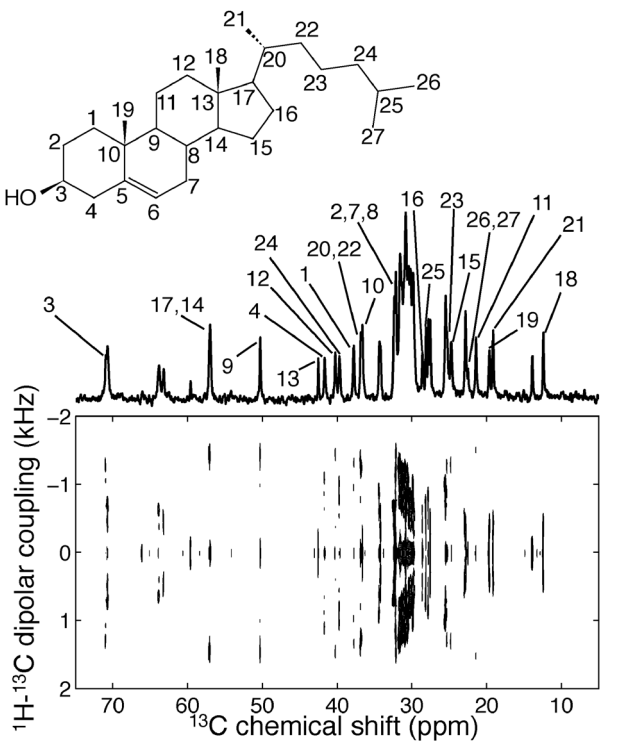


Fig. 4 2D ^1H–^{13}C R-PDLF spectrum of fully hydrated POPC–cholesterol multi-lamellar vesicles with 60 mol% cholesterol at 300 K and B_0 = 11.74 T with CP polarization transfer, 5 kHz magic-angle spinning, 50 kHz TPPM ^1H decoupling during ^{13}C acquisition, and 45 kHz R18$_7^7$ ^1H–^{13}C dipolar recoupling in the indirect dimension. The contact time for the CP transfer was 700 μs. A ^{13}C CP 1D spectrum is also shown with assignments of the cholesterol peaks. The assignment was based on the publication by Soubias *et al.*[53]

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segments of cholesterol, with the exception of C_6, from samples with natural abundance of isotopes. The assignment of the sterol peaks was based on a previous report by Soubias *et al.*[53] Previous measurements for cholesterol order parameters were restricted to C–H segments that may be selectively deuterated, such as C_2–C_7, C_{22} and C_{24}–C_{26}.[54–57]

Cholesterol effect on the phospholipid order parameters

To measure the cholesterol effect on the C–H order parameters of POPC, a combination of R-PDLF experiments with CP and INEPT as polarization transfer methods was used. The molecular dynamics and molecular structure in the membrane determine the intensities obtained from INEPT and CP experiments. While INEPT gives higher signal intensities than CP for spin systems with faster dynamics and low order parameters, as the dynamics slow down and/or the order parameters increase, there is a buildup of the CP intensities and concomitant decrease of the INEPT efficiency. A theoretical analysis of this effect may be found elsewhere.[52] To determine $|S_{CH}|$ for the different C–H segments one may then choose to use two R-PDLF versions to measure order parameters from more dynamical segments (INEPT version) and from more rigid segments (CP version).

The effect of cholesterol on the C–H order parameters of the *sn*-1 (palmitoyl), *sn*-2 (oleoyl) and headgroup segments is shown in Fig. 5, as well as cholesterol C–H order parameters for the ring and tail parts of the sterol, from both MD simulations and NMR experiments. Note that again, individual values for the separate C–H bonds in CH_2 groups are always shown for simulations (red dots). Experimental values show only one order parameter value from the methylene groups (with exception of the segments g_1 and C_2 at the *sn*-2 chain of POPC). This observation results from the fact that the two distinct C–H pairs in the CH_2 group have the same motional average, $\langle \cos^2 \theta \rangle$, which is the case for phospholipid acyl chains, or that only one of the C–H pairs have appreciable efficiency of polarization transfer, which applies to the cholesterol ring structure.

Cholesterol induces an increase of the population of *trans* conformers in the *sn*-1 and *sn*-2 acyl chains of the phospholipid, and hence induces an increase of the $|S_{CH}|$ values from

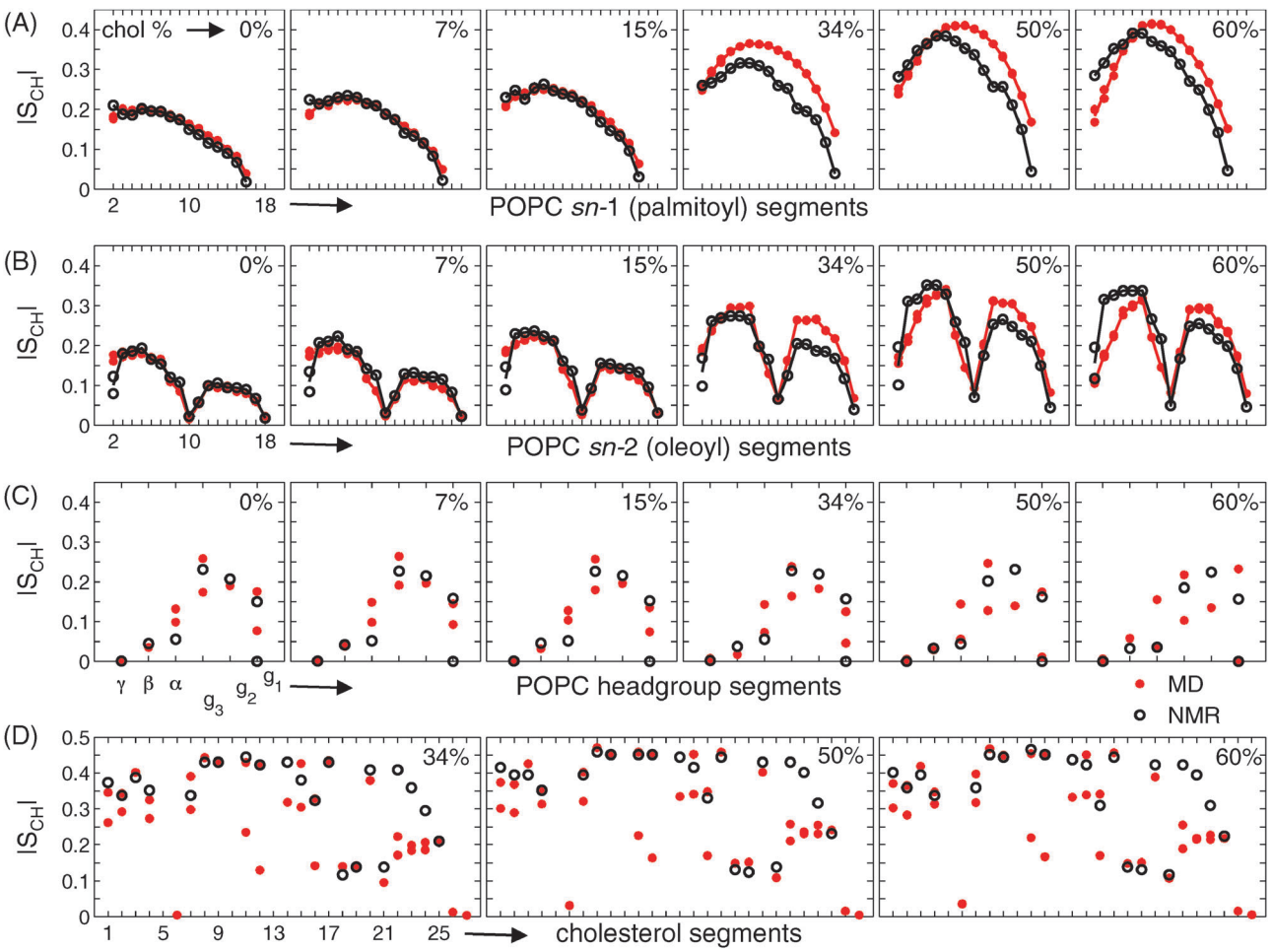


Fig. 5 Order parameter magnitude $|S_{CH}|$ *vs.* carbon segment number for the POPC *sn*-1 (palmitoyl), *sn*-2 (oleoyl) and headgroup chains (A, B and C respectively); as well as for the cholesterol ring and alkyl chain (D); in fully hydrated POPC–cholesterol bilayers at 300 K and a series of cholesterol concentrations. The $|S_{CH}|$ values were determined by ^1H–^{13}C solid-state NMR spectroscopy (black) and MD simulations (red). The solid lines in (A) and (B) follow the average values for each CH_2 segment. Note that, when it comes to simulations, we always show the individual order parameters from the different C–H bonds in a CH_2 group, while the NMR experiments often only yield the largest of the two values.

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the hydrophobic region. This effect is seen from both MD simulations and experiments. On the other hand, the experimental $|S_{CH}|$ values for the polar headgroup are rather unaffected by cholesterol. This experimental observation is not captured in the MD simulations that predict $|S_{CH}|$ values for the g_1, g_2, g_3, α and β segments which generally vary with respect to the cholesterol concentration. Moreover, all the experimental order parameters from the polar part of POPC are single valued for each carbon, with the exception of segment g_1, while the MD simulations predict two distinct order parameters for carbons g_1, g_3 and α.

Discussion

The pure phospholipid bilayer structure: MD *vs.* NMR

Hydrophobic region. Order parameters for acyl chains in pure phospholipid bilayers have been extensively studied by means of NMR experiments and simulations. The first ^{2}H NMR measurements were done in the 70's and since then the techniques have been widely used on these systems. Less publications exist with order parameters determined by ^{1}H–^{13}C NMR. A NMR database for both these types of measurements in lipid bilayers has been reported recently.[58] Generally, the $|S_{CH}|$ values obtained in different experimental studies are in good agreement and MD simulations reproduce these results well. This observation applies also to our experimental and NMR results as seen in Fig. 3. Slight deviations between our experimental $|S_{CH}|$ values and the previous ^{2}H NMR $|S_{CD}|$ values might be due to experimental imperfections, such as amplitude missettings and inhomogeneities in the ^{1}H rf field that may lead to 5–10% error in the R-PDLF NMR measurements.[59] The good agreement between our experimental order parameter profiles and previous experimental measurements indicates that our ^{13}C peak assignment is reasonable.

Only the simulated order parameters for carbon C_2 of the *sn*-2 chain deviate considerably from the experimentally measured ones (Fig. 3B). R-PDLF NMR gives two distinct C–H order parameters for C_2 with lower values than the single order parameter value from the $C_{2'}$ methylene at the *sn*-1 chain (triple splitting pattern around 34 ppm in Fig. 2). This feature has also been systematically observed from phospholipids other than POPC with ^{2}H NMR experiments.[60] It has previously been shown that the Berger force-field, which is used in this work, gives equal order parameters for C_2 and $C_{2'}$[61] but that other force-fields such as GAFF[61] and CHARMM C36[62] predict different values. However, these last force-fields also do not give distinct values for the different C–H bonds of the C_2 carbon in the *sn*-2 chain. It is known that the separate order parameters for C_2 are due to different motional averaging of the two C–H bonds,[63] *i.e.* the different bonds point on average in different directions with respect to the bilayer normal. Moreover, this difference is most likely due to different orientations of the C_1–C_2 bonds at the *sn*-1 (perpendicular to the membrane plane) and the *sn*-2 (parallel to the membrane plane) chains according to X-ray studies.[64,65] A correct prediction of these distinct C–H bond alignments by simulation models is most likely important for a proper description of the intermolecular interactions with the *sn*-2-carbonyl.

Table 1 Comparison between headgroup order parameters from different phospholipid bilayers

Carbon label	$\|S_{CH}\|$[a] POPC	S_{CH}[b] DMPC	$\|S_{CD}\|$[c] DPPC	$\|S_{CD}\|$[d] *E. coli*
g_1	0.00	0.00	0.00	—
	0.15	−0.15	0.13	—
g_2	0.21	−0.20	0.20	—
g_3	0.23	−0.23	0.23	0.20
	—	—	0.21	0.21
α	0.06	+0.04	0.05	0.08
β	0.04	−0.03	0.04	0.03
γ	0.00	0.00	0.01	0.00

[a] $|S_{CH}|$ from POPC MLVs at 300 K. [b] S_{CH} measured by Gross *et al.*[67] for DMPC at 303 K. [c] $|S_{CD}|$ from ^{2}H NMR on DPPC bilayers at 322 K published by Seelig and co-workers.[66] [d] $|S_{CD}|$ from ^{2}H NMR on *E. coli* phospholipids at 310 K by Gally *et al.*[68] Note that all columns show the absolute values of the order parameters except one, from Gross *et al.*,[67] who also measured the sign of the order parameters.

Choline headgroup and glycerol backbone. The order parameters of the headgroup and the glycerol backbone in other phospholipids than POPC have been investigated by means of ^{2}H NMR.[66–68] Previously published order parameters are shown in Table 1 together with our results. From the table we see that our results are very similar to the ones measured from phospholipid membranes containing dipalmitoylphosphatidylcholine (DPPC), dimyristoylphosphatidylcholine (DMPC) and lipids extracted from *E. coli.* As previously suggested by Gally *et al.*,[68] the similarity of the polar headgroup and glycerol backbone configuration in the four systems indicates that the general behavior of these groups is independent of the molecular structure in the hydrophobic region.

Table 1 shows that the two CH_2 segments in the glycerol backbone, C_{g1} and C_{g3}, have two distinct order parameters each. These differences are due to a motional non-equivalence of the individual C–H bonds of these segments. For C_{g1}, the difference between the two distinct C–H order parameters is large, with one of the C–H bonds retaining an angle close to the magic angle relative to the motional axis, and thus giving an order parameter very close to zero.[67]

On the other hand, the difference between the distinct $|S_{CH}|$ values from C_{g3} is very small. Although we could not measure such a slight difference because of insufficient resolution in the ^{1}H–^{13}C dipolar coupling dimension of the R-PDLF experiment, it has been detected in previous ^{2}H NMR studies.[66,68]

Other types of anisotropic NMR interactions have been measured for the glycerol and choline moieties, *e.g.*, the ^{31}P chemical shift anisotropy (CSA) and a number of ^{31}P–^{1}H and ^{31}P–^{13}C dipolar couplings.[69] A large set of NMR observables available may be used as constraints to find an average structure for the polar group. The first detailed model for the structure of the choline headgroup and glycerol backbone was built based on ^{2}H and ^{31}P NMR results from DPPC bilayers and crystalography studies.[66,70] This model assumes rapid transitions between two enantiomeric choline states, a free rotation around the C_{g1}–C_{g2}–C_{g3}–O dihedral, and the assumption that the C_{g2}–C_{g3} bond is on average perpendicular to the plane of the bilayer.[66] Such description captures almost all

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NMR parameters; however, the last two assumptions are not compatible with the two distinct order parameters for C_{g3}, and thus other models were proposed *e.g.* in which the angle of the C_{g1}–C_{g2}–C_{g3}–O dihedral was completely fixed.[69] In general, all the different models proposed share a strong similarity with the glycerol–choline headgroup structure of the DMPC-B conformer identified in DMPC solid crystals by Pearson and Pascher,[70] and a common feature in all the NMR-based descriptions is that the P–N electric dipole is oriented almost parallel to the plane of the membrane.

MD simulations would be a reasonable method for trying to find the more realistic description over the set of different models. However, the order parameters of the polar part of POPC calculated from our MD simulation model are rather different than the experimental values as seen in Fig. 3C, indicating that the used simulation model cannot describe the structure of the glycerol and choline groups with a high level of detail. Despite the experimental values available, this kind of comparisons to MD simulations have been rare. Only recently it was shown that the careful refinement of torsional parameters improved the all-atom CHARMM model for the polar headgroups.[62] Correct modeling of the choline headgroup and glycerol backbone orientation is highly important. For instance, a different orientation of the P–N dipole may lead to a different electrostatic potential around the membrane.

Cholesterol effect on the bilayer structure

Ordering of the acyl chains. In Fig. 5A and B we see a systematic increase in the acyl chain order parameters for both *sn*-1 and *sn*-2 chains when the cholesterol concentration increases. This effect, linked to condensation upon addition of cholesterol,[71,72] has been investigated by MD simulations and ^{2}H NMR for a large number of systems due to its possible implications on *lipid raft* formation and/or protein function regulation.[14,15] Also a ^{1}H–^{13}C recoupling NMR study on the effect of cholesterol on DOPC bilayers has been reported previously.[73] However, systematic and quantitative comparison between simulations and experiments has not been presented for the dependency of the complete acyl chain order parameter profile on cholesterol concentration.

At cholesterol molar fractions below 15%, Fig. 5A and B show a good agreement between measured and simulated acyl chain order parameters for both chains. The agreement is not as good for higher cholesterol concentrations, where the simulation model gives an overestimate of the ordering, mostly to the methylenes towards the center of the bilayer. Conversely, the order of carbons closer to the headgroup is underestimated at the highest concentrations. The most obvious reason for this discrepancy would be an inaccurate force-field description. Especially inter-molecular van der Waals interactions are difficult to parametrize and would be a plausible reason, although we removed one common feature that leads to over-condensation (we have changed CH_2/CH_3 groups in cholesterol to LP_2/LP_3 groups to make it more consistent with the Berger parameters).[42] Several other reasons related to the simulation force-field could explain the differences, such as the incorrect orientation of the C_1–C_2 bond at the *sn*-2 chain or a wrong positioning of cholesterol molecules in the bilayer. Moreover, we cannot exclude also that the simulation might be too small in number of lipid molecules (128) and/or too short in time length (100 ns), since it is not clear from experimental studies if a change in phase morphology occurs above 15 mol% cholesterol. This possibility is discussed in more detail in a separate section.

The acyl chain ordering is related to the number of *gauche* and *trans* configurations along the chain, which in turn relates to how much the chain is stretched, *e.g.* the maximum stretch of the chain corresponds to an all-*trans* conformation. A number of models based on statistical mechanics have been proposed to describe this feature quantitatively. Such models provide relations between S_{CH} values of saturated acyl chains and methylene travel values,

$$\langle D_i \rangle = \langle z_{i-1} \rangle - \langle z_{i+1} \rangle, \quad (4)$$

where $\langle D_i \rangle$ is the average distance between two carbons, C_{i+1} and C_{i-1} bonded to carbon C_i, projected onto the bilayer normal.[74–76]

Fig. 6A shows how well two of these models, namely the *diamond lattice* (DL)[75] and the *mean torque* (MT)[74] models, reproduce our MD data. The explicit equations for the relations $S_{CH} \rightarrow \langle D_i \rangle$ can both be found in ref. 74. Since the MT relation seems to give a better prediction of the values $\langle D_i \rangle$ for our MD simulations, we use this model to convert the ^{1}H–^{13}C NMR

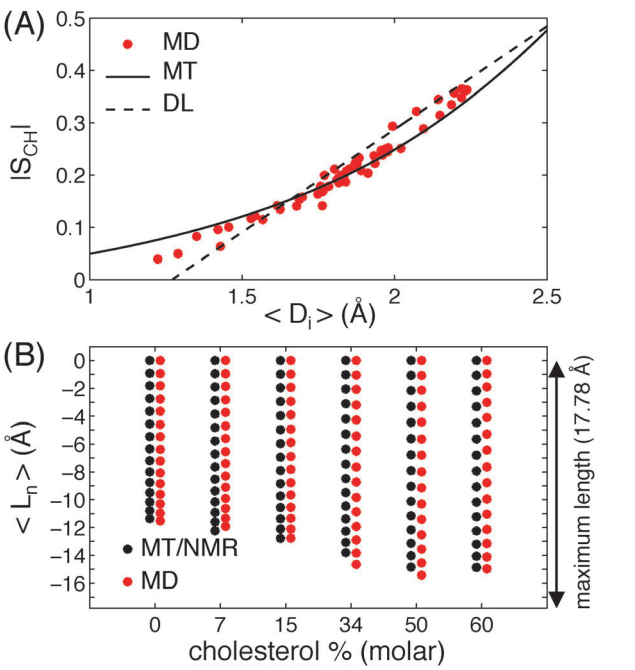


Fig. 6 (A) Acyl chain order parameter $|S_{CH}|$ *vs.* methylene travel $\langle D_i \rangle$ calculated with the mean torque (MT, solid line) and diamond lattice (DL, dashed line) models, as well as estimated directly from the MD simulations (red dots) for the *sn*-1 (palmitoyl) chains of all simulated systems. (B) Projected length in the bilayer normal of the *sn*-1 chain from MD simulations and by using the MT model[74] on ^{1}H–^{13}C NMR order parameters from fully hydrated POPC–cholesterol mixtures at 300 K.

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order parameters of the POPC *sn*-1 chain to $\langle D_i \rangle$ values. Here, the projected length from carbon 2 to carbon n is defined as

$$\langle L_n \rangle = 1/2 \sum_{i=3}^{n} \langle D_i \rangle - D_n/2, \qquad (5)$$

Fig. 6B shows how the projected lengths of the *sn*-1 (palmitoyl) chain change as function of cholesterol concentration. The NMR data together with the MT model suggest that the *sn*-1 chain will stretch, with increasing cholesterol concentration, until it reaches a plateau value at around 50 mol% cholesterol. The directly simulated distance $\langle z_{16} \rangle - \langle z_2 \rangle$ and the total projected length determined using the experimental order parameters and the MT model of Petrache *et al.*[74] agree within an error of 1 Å. This indicates that even though a discrepancy between the $|S_{CH}|$ values from MD simulations and NMR experiments is found above 15 mol% of cholesterol, the length of the extension of the acyl chains in the MD simulations might be very close to the real one.

Effect on the structure and dynamics of the glycerol backbone and choline headgroup. A very small increase of the order parameters for segment C_{g2} and a more significant decrease for segment C_{g3} are seen with increasing cholesterol concentration, particularly at 50 and 60 mol% cholesterol, whereas segment C_{g1} maintains a constant order parameter through the entire range of cholesterol concentrations studied. It is also evident from the data presented in Fig. 5C that while the values for the order parameters for C_{g2} and C_{g3} maintain a fixed pattern relative to each other up to 34 mol% cholesterol, this trend changes at and above 50 mol% cholesterol. Equal order parameters for these segments would be expected to result from a parallel orientation of the C_{g2}–C_{g3} bond with respect to the bilayer normal.

Thus, we may interpret the unequal trends on C_{g2} and C_{g3} order parameters as a slight change of either the *gauche*/*trans* conformational distribution around the C_{g1}–C_{g2} bond, that may change the C_{g2}–C_{g3} bond orientation, and/or due to a possible change of the *gauche*/*trans* conformational distribution around the C_{g2}–C_{g3} bond. The variations in the $|S_{CH}|$ values of C_{g2} and C_{g3} may be a consequence of an increase in conformational entropy of the polar headgroup at the higher (50 and 60 mol%) cholesterol concentrations, *i.e.* an increase in free space available at the hydrophilic region.

For the choline group, a very modest decrease is seen for the C_α and C_β order parameters. This observation is in agreement with previous ^{2}H NMR studies on DPPC–cholesterol bilayers.[77] These changes can be explained by a slight change of 2.5° in the average dihedral angle N–C–C–O.[77] Moreover, a redistribution of conformational states usually leads to a variation of the ^{13}C chemical shifts (ref. 78 and references therein). The chemical shifts from the choline and glycerol carbons are essentially unchanged with the highest variation being 0.1 ppm for g_1, which is quite small compared to variations of about 1.5 ppm observed in the acyl chain groups. This indicates that unlike the *sn*-1 and *sn*-2 chains, the conformations of the glycerol and choline groups are rather unchanged.

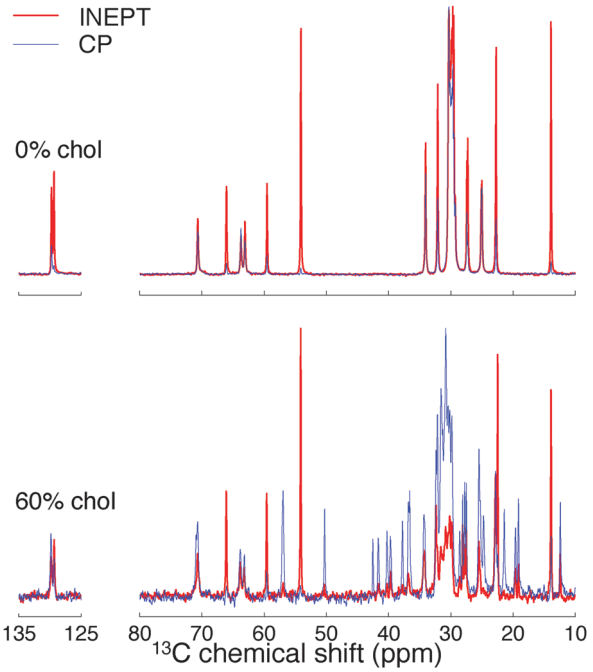


Fig. 7 Comparison of ^{13}C CP (blue) and refocused INEPT (red) spectra of fully hydrated multi-lamellar vesicles with different POPC–cholesterol compositions recorded at 300 K and B_0 = 11.74 T with 5 kHz magic-angle spinning and 50 kHz TPPM ^{1}H decoupling. The CP contact time was 700 μs. All four spectra were recorded with accumulation of 256 transients.

Qualitative information about C–H bond reorientation dynamics can be obtained from the ratios between the CP and INEPT peak intensities, which depend on both the order parameter and correlation time for reorientation.[52] CP and INEPT spectras are shown in Fig. 7 where ^{13}C peaks can be seen for the choline headgroup and glycerol backbone measured from POPC membranes with and without cholesterol. For choline headgroup carbons (54.1, 59.6 and 66.1 ppm) the CP/INEPT peak intensity ratio is low and does not change, but for the glycerol backbone carbons (63.1, 63.7 and 70.7 ppm) the CP signal is slightly enhanced when cholesterol is added. Since the order parameters are essentially unchanged for all the polar groups, we interpret the different CP/INEPT peak intensities as a slight slow down of the glycerol backbone dynamics due to incorporation of cholesterol and a high flexibility of the choline group which is not affected by cholesterol concentration.

Altogether, the present NMR results suggest that the choline headgroup order and dynamics are not significantly affected by cholesterol, while a very small effect is observed in the glycerol backbone. This observation is in good agreement with previous ^{2}H-NMR data on DPPC[77] and T_1(^{14}N) NMR relaxation studies.[79] Such small variations in the order parameters and dynamics of the glycerol backbone obtained in this work are not reproduced by the MD simulations. Therefore, models for the effect of cholesterol on the headgroup structure based on the MD simulations using the present GROMOS and Berger force-fields or other force-fields lacking previous comparison with NMR experiments, as *e.g.* in ref. 80 and 81, may be misleading since the MD simulation outcome is unrealistic at this level of detail.

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Cholesterol orientation in the bilayer

There is good agreement between simulations and experiments for the ring structure of cholesterol, except for C_{14} as shown in Fig. 5D. The discrepancy for C_{14} is surprising since all the other values in the rigid ring seem rather reasonable. In respect to the cholesterol alkyl chain, we observe clearly lower $|S_{CH}|$ values from simulations compared to the experimental ones for three carbons, C_{22}–C_{23}–C_{24}. Based on these observations, we suggest that while the behavior of the sterol rigid structure is well described by the simulation model, the cholesterol tail is either too flexible or incorrectly oriented. Just recently, Lim and coworkers have proposed a new CHARMM force-field for cholesterol which gives perfect agreement with C–D order parameters for the C_{22} and C_{24}–C_{26} cholesterol segments in DMPC bilayers.[82]

Since the rigid structure of the cholesterol ring system is known, a single value measured from any of the rings is enough to determine the average angle between the director axis of the cholesterol molecule and the membrane normal.[54,83] The precise orientation distribution of cholesterol is however not possible to be determined from these data alone, *i.e.* the same order parameters can be measured from a molecular axis vector fixed to a certain angle or fluctuating around it.[84] Below, we show how to determine the different possible orientation distributions from our NMR results and how MD simulations are specially useful to interpret the NMR measurements in this case.

In the traditional NMR approach, the order parameter for the director axis of the cholesterol molecule with respect to the bilayer normal, $S_\alpha = \frac{1}{2}\langle 3\cos^2\alpha - 1\rangle$, where α stands for the angle between the molecular director and the bilayer normal, is determined by using the equation

$$S_{CH} = \tfrac{1}{2}\langle 3\cos^2\alpha - 1\rangle \; \tfrac{1}{2}\langle 3\cos^2\beta - 1\rangle = S_\alpha S_\beta, \qquad (6)$$

where S_{CH} is the order parameter measured by NMR and S_β is the order parameter of the C–H bond relative to the molecular director which may be determined by knowing the rigid cholesterol structure.[83] This equation applies for any C–H bond in the rigid cholesterol ring. S_α can arise from several possible angle distributions, $f(\alpha)$, which satisfy the equation

$$S_\alpha = \frac{\int_0^\pi f(\alpha)\frac{1}{2}(3\cos^2\alpha - 1)\mathrm{d}\alpha}{\int_0^\pi f(\alpha)\mathrm{d}\alpha}, \qquad (7)$$

and the average angles for the different distributions may then be determined by

$$\langle\alpha\rangle = \frac{\int_0^\pi \alpha f(\alpha)\mathrm{d}\alpha}{\int_0^\pi f(\alpha)\mathrm{d}\alpha}. \qquad (8)$$

Generally, normal distribution functions are assumed,

$$f(\alpha) = \sin\alpha \mathrm{e}^{-\frac{(\alpha-\alpha_0)^2}{\sigma^2}}, \qquad (9)$$

where sin α accounts for the equal probability of the different orientations in a sphere. The gaussian term can be interpreted such that the cholesterol fluctuates around an angle α_0 with respect to the membrane normal, and an amplitude of fluctuations determined by σ. The assumption that these fluctuations have a gaussian form has also been used to analyze data from other rigid molecules such as pyrene or WALP23.[84–86] Since by using eqn (6) we may calculate S_α, it is possible then to determine which α_0 and σ values satisfy eqn (7).[83] Note that with the analysis above we get only a set of possible values for α_0, σ and $\langle\alpha\rangle$. On the other hand, the distribution for α can also be directly calculated numerically from molecular dynamics simulations.

We show in Fig. 8 such type of analysis for the cholesterol tilt in POPC–cholesterol bilayers containing 34 mol% cholesterol (see the ESI† for the explicit procedure). Fig. 8A shows that the average angle can be determined relatively unambiguously but the distribution can be something between two extreme cases: a tilt angle fixed at 17° or fluctuating in the range 0°–40°. By comparing the possible gaussian distributions, from the set of solutions of eqn (7), with the numerical distribution obtained from the MD simulation (Fig. 8B, red), we find that the best analytical function to describe that the orientation

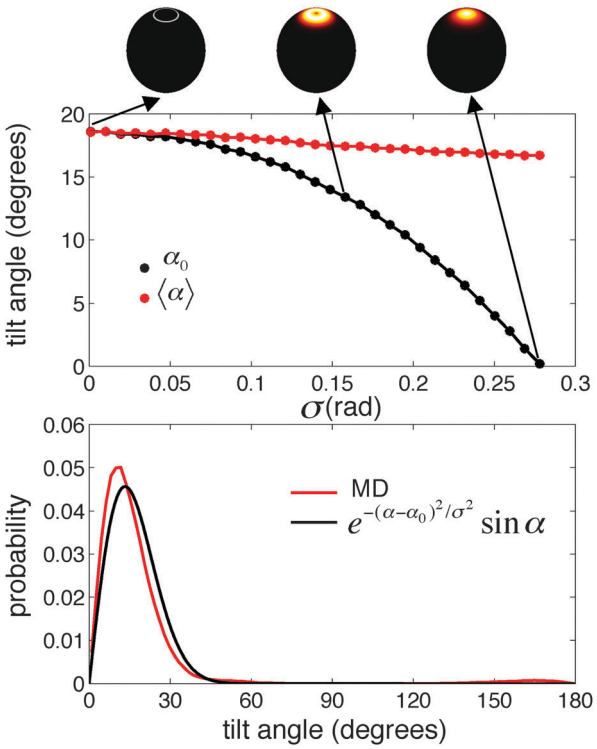


Fig. 8 (A) Possible gaussian distributions of the orientation of cholesterol in POPC–cholesterol MLVs (34 mol% cholesterol) at 300 K, based on the ^1H–^{13}C NMR dipolar coupling of the carbon bound to the hydroxyl group (C_3). For a particular distribution (σ and α_0, black), the standard deviation σ defines the range of angles that the molecular director of cholesterol ($C_{17}\rightarrow C_3$) may adopt relative to the bilayer normal and α_0 stands for the most energetically favorable angle. The average tilt angle for each possible distribution is also shown (red), as well as spheres mapping the probability of different orientations (dark to white is from lower to higher probability) for three distinct gaussian distributions. (B) Cholesterol tilt angle distributions in POPC–cholesterol bilayers (34 mol% cholesterol, 300 K) from MD simulations (red) and the gaussian distribution (black) which best fits the MD simulation: $\alpha_0 = 0°$ with a standard deviation of $\sigma = 16°$.

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distribution has $\alpha_0 = 0^\circ$ with standard deviation of $\sigma = 16^\circ$ and average angle $\langle\alpha\rangle = 17^\circ$, *i.e.* the tilt angle fluctuates between 0°–40° and the most energetically favorable orientation is when cholesterol is parallel to the bilayer normal.

Is there phase heterogeneity in the POPC–cholesterol system?

As discussed above, there is excellent agreement between acyl chain order parameters of POPC measured by NMR and calculated by MD simulations for concentrations of cholesterol up to 15 mol%. Above this value the simulated and experimentally measured order parameters diverge. Possible explanations regarding force-field issues have been discussed above. Another possibility is that some kind of lateral heterogeneity occurs for the POPC–cholesterol mixture above a certain cholesterol content. Since the initial configurations for our MD simulations were laterally homogeneous, the simulations of 100 ns may be too short and/or the size of the simulated systems too small to reach lateral heterogeneity. A liquid disordered/liquid ordered (L_d/L_o) phase coexistence has been suggested from fluorescence studies at 300 K (the temperature at which the present work was done) between 8 and 40 mol%[87] or between 12 and 45 mol%[88] cholesterol in the bilayers. Macroscopic phase separation into L_d and L_o would be detected in the R-PDLF experiment as the presence of both lower and higher distinct order parameters for each C–H bond. Just as in an earlier ^{2}H NMR study,[89] there are no indications of more than one order parameter per C–H bond for the fully hydrated POPC–cholesterol MLVs at 300 K. As justified in the ESI,† these observations show that if there is L_d/L_o separation, molecular exchange between the two phases must occur on a time scale much smaller than 10^{-5} s, corresponding to a lateral domain size much smaller than 100 nm.

Although the L_d/L_o coexistence[71] is generally accepted for binary mixtures of cholesterol and phospholipids with saturated acyl chains, such as DPPC[90,91] or sphingomyelin,[92] the existence of a phase separation in bilayers prepared from binary mixtures of cholesterol with phospholipids that have one saturated and one unsaturated acyl chain is not universally accepted and has been contested by various laboratories using very different methods: on the basis of in-plane translational diffusion studies without use of invasive probes,[93] from dilatometric studies (also without use of invasive probes),[94] from studies on the kinetics and thermodynamics of insertion of amphiphiles into lipid bilayers,[24] from the visualization of fluorescent membrane domains in giant lipid vesicles,[95] and most recently from a combination of several techniques[96] and from spin-label EPR studies.[97] Alternatively, 2D micelles formation in lipid bilayers prepared from binary mixtures of cholesterol and phospholipids with one saturated and one cis-unsaturated acyl chain, such as POPC, has been proposed, in which cholesterol represents the 2D core with the POPC molecules oriented such that the saturated acyl chain faces the sterol and the unsaturated point towards the outer part of the 2D micelle.[24] Since the components of the putative micelles would be in rapid equilibrium with the bulk lipid phase of the bilayers, this would be a case of microscopic heterogeneity, thermodynamically quite distinct from a phase separation. Due to the fast dynamics of micellization, this type of heterogeneity would not be detectable in our NMR experiments and the POPC–cholesterol bilayer would appear to be a single homogeneous phase.

We can neither rule out nor confirm the presence of any type of lateral heterogeneous domains smaller than 100 nm. Thus, it is not clear if the deviation between the $|S_{CH}|$ values from simulations and experiments is related to: (I) the onset of a certain phase transition, which could not be captured by the simulations due to the computational restrictions on system size and/or number of time steps performed; (II) simply because the force-field parametrization is not accurate enough to predict the effect of cholesterol on the acyl chains at this level of detail.

Conclusion

R-PDLF NMR, with CP and INEPT as alternate polarization transfer methods, is useful to study POPC bilayers which contain cholesterol since it provides a large set of detailed data on the molecular structure of both the phospholipid and cholesterol. The experimental data obtained suggest that, while acyl chains adopt more *trans* configurations, the behavior of the glycerol backbone and the choline headgroup of POPC is rather independent of the cholesterol concentration.

We have shown that a widely used MD simulation model for phospholipid–cholesterol mixtures reproduces well the ordering effect of cholesterol upon the POPC acyl chains at cholesterol concentrations up to 15 mol% but deviates from the experimentally observed NMR results above this concentration. The cholesterol ring structure is also well reproduced at all cholesterol concentrations examined.

The unprecedented detail of the experimental data allows us to point out four main deficiencies of the MD simulation model:

(I) The experimental order parameters for the POPC choline headgroup and glycerol backbone and their qualitative behavior as a function of cholesterol are not correctly predicted by the used simulation model. The development of a more accurate model of the phospholipid headgroup is most likely important since such type of models are often used to study detailed interactions between the phospholipid headgroup and other molecules, like cholesterol or proteins.[17,98]

(II) Acyl chain order parameters are not well reproduced with cholesterol concentrations of 34 mol% cholesterol and higher. It is not clear though if this deviation arises from problems in the force-field or from lateral heterogeneity in the bilayer which could not be captured in the present simulations.

(III) The experimental order parameters of the cholesterol alkyl chain are not reproduced by simulations, meaning an incorrectly oriented or too flexible alkyl chain. The most likely reason is that the dihedral angle parameters for the alkyl chain are not properly defined in this model. A better description of the cholesterol alkyl chain is important for interactions between cholesterol and other molecules at the center of the bilayers.

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(IV) The distinct POPC experimental order parameters for segment C_2 for each acyl chain may be interpreted to be due to a different orientation of the C_1–C_2 bond of the *sn*-2 chain (parallel to the bilayer plane) compared to the orientation of the $C_{1'}$–$C_{2'}$ at the *sn*-1 chain (parallel to the bilayer normal). This difference is not captured by the simulation model and may be important for describing phospholipid–cholesterol interactions.

The main relevance of this work is that it has been possible to measure a vast number of C–H order parameters, from both POPC and cholesterol molecules, that had not been previously measured by ^{2}H NMR methods. We believe that the experimental data presented here may be an important contribution to the development of more realistic force-fields for MD simulations.

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